

Zainab Tairu

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Bio

I am a Machine Learning Engineer specializing in NLP and AI accessibility, Passionate about improving speech to text systems, rag and creating solutions for low resource languages and knowledge driven AI

Skills

Hard Skills

- Deep Learning** Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs)
- NLP** Named Entity Recognition (NER), Sentiment Analysis, Transformer Models, Rasa, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs).
- Computer Vision** Image Classification
- Languages** Python, SQL, Cypher
- Mathematical Analysis** Linear Algebra, Calculus, Probability and Statistics
- Libraries** Pytorch, spaCy, NLTK, Sckit-Learn, HuggingFace, Tensorflow
- Web Frameworks** FastAPI
- Version Control** Git, GitHub

Soft Skills

Communication, Teamwork, Problem-Solving, Leadership, Adaptability, Creativity, Presentation, Attention to Detail, Time Management.

Experience

- The Innoverse (Sandton, South Africa)** Oct 2024 – Present
NLP intern
 - Led the design, development, and launch of the company's WordPress website, improving user experience and boosting brand visibility.
 - Designed an interactive chatbot aimed at streamlining customer service interactions, pending deployment.
 - Designed a knowledge graph to improve the response accuracy and user engagement of the chatbot.
 - Integrated Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) in the chatbot to significantly improve response quality and personalization.

- Hamoye (Lagos, Nigeria)** Nov 2023 – Dec 2023
Intern
 - Worked under the guidance of industry experts to train a predictive model to assess the impacts of climate change on Agricultural products
 - Wrote a research proposal for the Prediction of African conflicts advocating for proactive strategies based on our analytical insights to guide crucial policy decisions.
 - Designed a PowerPoint presentation highlighting the potential risks of food-related conflicts in African nations to a panel of judges.

Publications (In Progress)

- Multi-attention mechanisms with Transformer based NER models for Resume Parsing** Dec 2023 – Present
Co-authored with Dr. Avashna Govender
 - Co-authored a research paper on improving Named Entity Recognition (NER) for resume parsing using multi-attention mechanisms with transformer-based models.
 - Developed an innovative multi-attention mechanism that significantly enhanced the model's ability to identify and contextualize key resume elements such as skills, certifications, and educational background.

Personal Projects

- LET ME COOK** Nov 2024 – Present
 - Originated and refined a unique dataset of a diverse collection of recipes adapted to appeal to the Nigerian palate with a specific aim to empower students to efficiently utilize their pantry items here

Education

Bachelor of Science in Chemical Engineering
Obafemi Awolowo University, Osun, Nigeria
Expected Graduation: August 2026

Jan 2022 – Present

Volunteering Work

Star Girls Africa
Data science tutor

Jun 2024 – October 2024

- Authored a comprehensive beginner’s curriculum in data science.
- Delivered training to 100 students, enhancing their foundational skills.
- Achieved notable increase in student engagement and skills proficiency

Hackathon

SPE Hackathon — Osun State, Nigeria, January 2025

- Developed a machine learning model tp predict porosity in reservoir characterization.
- Authored a Research paper on the methodology, challenges, and results of implementing machine learning for geological assessments in Nigeria’s oil sector

Conferences/Workshops

Deep Learning Indaba

Location: Dakar, Senegal **Date:** September, 2024

- Participated in workshops and sessions on AI ethics, transformer models, and low-resource NLP, gaining hands-on experience with cutting-edge machine learning techniques.

Data Science Network AI Bootcamp

Location: Lagos, Nigeria **Date:** October 2024

- Participated in a hackathon to develop an English-to-Yoruba translation model, addressing NLP challenges for low-resource languages.

References

Dr. Avashna Govender
Research Group Leader, CSIR
✉ avashna002@gmail.com

Mr. Simphiwe Zitha
Deep Learning Indaba Mentor
✉ nhlanhla34@gmail.com